



CSE302: Database Systems (Section No. 4)

[Semester Year: Fall 2025]

Project Title:
Cricket League/Tournament Management

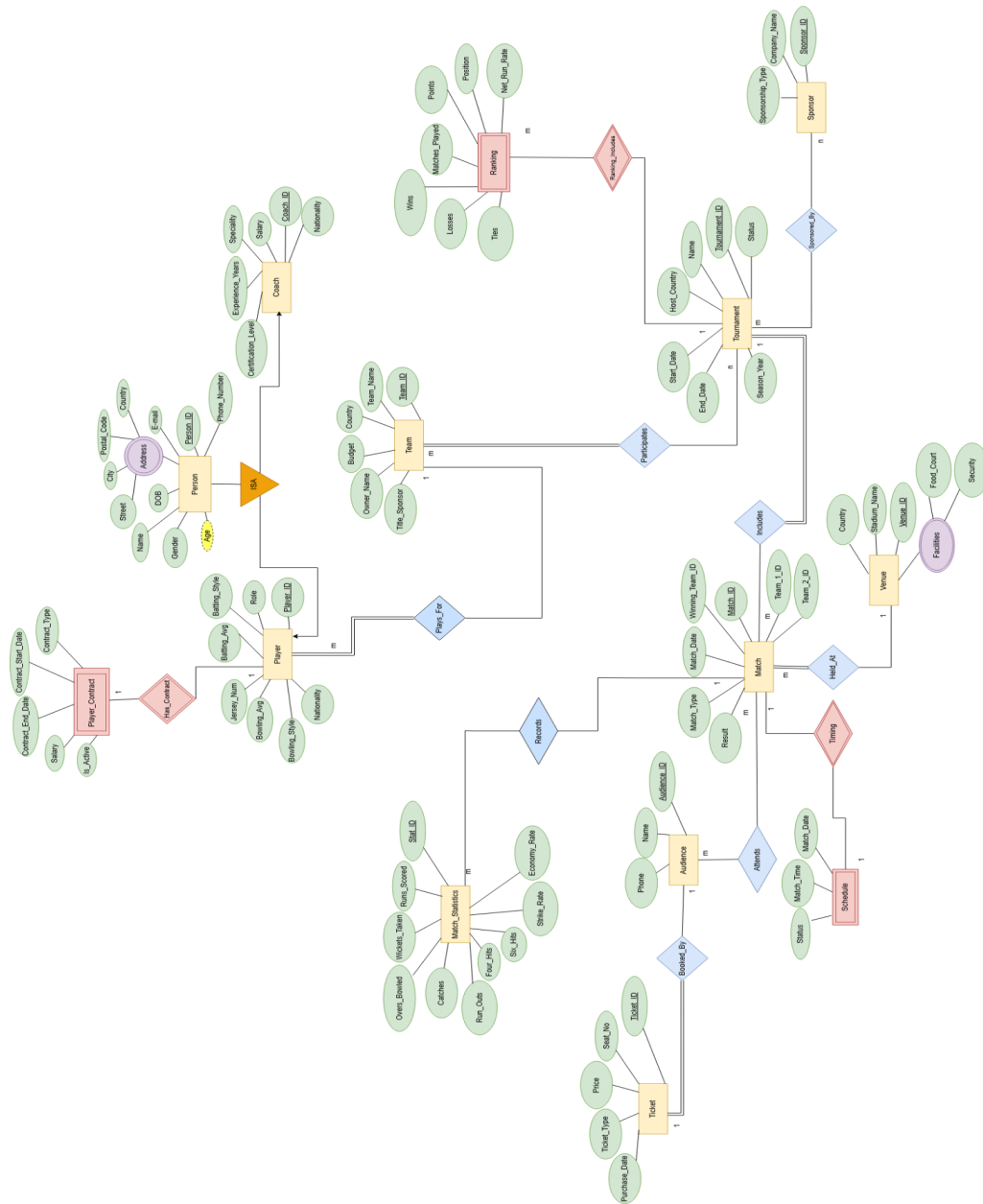
Project URL/Link:

<https://oraclepex.com/ords/r/ewu2025/cricket-tournament-management/>

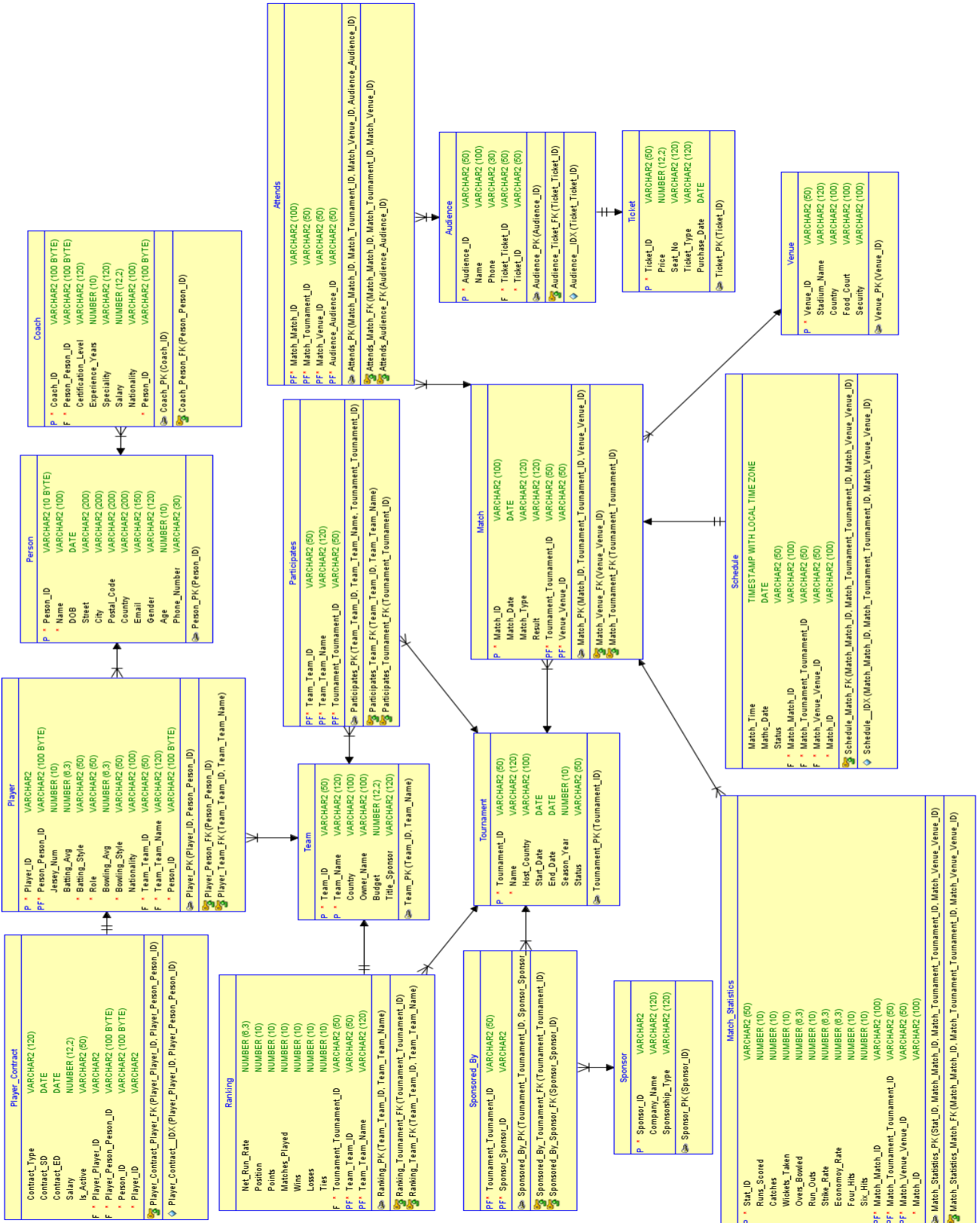
Submitted by:

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1. E-R Model of the Project



2. Schema Design of the Project



3. Implementation Checklist

Description	Remarks
Project ID	176948
Workspace Name	EWU2025
Workspace Username	Admin
Workspace Password	admin123
Authentication Type	Workspace Administrator
Username and Password to Login	
username and password for multiple users	
Admin	Admin123
Coach	Coach123
Player	Player123
User	user123
Number of Reports with Forms	17
Number of Reports based on Queries, Aggregate Queries	12
Included any trigger/other feature which is not taught in class. If yes, Please explain briefly.	Yes 1.trg_person_age - Automatically calculates and updates the derived attribute age from dob (date of birth) before inserting or updating a row into the PERSON table.
Write the most unique feature/functionality in your application or the page of your application that you are proud of. Explain briefly.	This feature set proves that our application is not just CRUD-based, but: ● Well-structured (de-redundant schema using surrogate primary keys) ● Automated (trigger-based age calculation from date of birth) ● Reliable (strong CHECK, UNIQUE, and FOREIGN KEY constraints) ● Interactive bar and donut charts to transform complex sports data into clear, visual performance insights.

	It mirrors a real-world sports management system where maintaining data integrity, automating processes, and managing relationships are crucial for smooth operation.
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4. Reports with Forms

Report Page Number and Name	Form Page Number and Name	Table
Page No – 2, Person Report	Page No – 3, Person Form	Person
Page No – 4, Team Report	Page No – 5, Team Form	Team
Page No – 6, Tournament Report	Page No – 7, Tournament Form	Tournament
Page No – 8, Venue Report	Page No – 9, Venue Form	Venue
Page No – 10, Ticket Report	Page No – 11, Ticket Form	Ticket
Page No – 12, Audience Report	Page No – 13, Audience Form	Audience
Page No – 14, Coach Report	Page No – 15, Coach Form	Coach
Page No – 16, Match Report	Page No – 17, Match Form	Match
Page No – 18, Schedule Report	Page No – 19, Schedule Form	Schedule
Page No – 20, Match Statistics Report	Page No – 21, Match Statistics Form	Match Statistics
Page No – 22, Attends Report	Page No – 23, Attends Form	Attends
Page No – 24, Participates Report	Page No – 25, Participates Form	Participates
Page No – 26, Ranking Report	Page No – 27, Ranking Form	Ranking
Page No – 28, Player Report	Page No – 29, PlayerForm	Player
Page No – 30, Player Contract Report	Page No – 31, Player Contract Form	Player Contract
Page No – 32, Sponsor Report	Page No – 33, Sponsor Form	Sponsor
Page No – 34, Sponsored By Report	Page No – 35, Sponsored By Form	Sponsored By

Screenshots (Reports with Forms):

Figure 4.1: User Report

Cricket Tournament Management						aura
Ticket Audience Coach Match Schedule Match Statistics Attends Participates	Users					
	Q Go Actions Create					
		User ID	User Name	User Pass	User Roll	User Activated
		1	Admin	Admin123	admin	1
		2	Coach	Coach123	coach	1
		3	Player	Player123	player	1
		4	User	user123	user	1
						1 - 4

Figure 4.2: Sponsored By

Cricket Tournament Management

aura

Ticket

Audience

Coach

Match

Schedule

Match Statistics

Attends

Participates

Sponsored By

Q

Go

Actions

Create

	Sponsor ID	Tournament ID
	SP001	TR001
	SP002	TR001
	SP003	TR002
	SP004	TR003
	SP005	TR004

1 - 5

Figure 4.3: Sponsor

Cricket Tournament Management

aura

Ticket

Audience

Coach

Match

Schedule

Match Statistics

Attends

Participates

Sponsor

Go

Actions

Create

	Sponsor ID	Company Name	Sponsorship Type
	SP001	Grameenphone	Title Sponsor
	SP002	Robi	Official Partner
	SP003	Bangladesh Bank	Financial Sponsor
	SP004	Walton	Equipment Sponsor
	SP005	Pran	Financial Sponsor

1 - 5

Figure 4.4: Player Contract

Cricket Tournament Management

Ticket

Audience

Coach

Match

Schedule

Match Statistics

Attends

Participates

Player Contract

Q

Go

Actions

Create

	Contract ID ↑	Player	Contract Type	Contract Start Date	Contract End Date	Salary	Is Active
	CT001	P001	Full-time	1/1/2023	12/31/2025	2000000	Yes
	CT002	P002	Full-time	1/1/2023	12/31/2024	1800000	Yes
	CT003	P003	Half Time	6/1/2023	5/31/2025	1500000	Yes
	CT004	P004	Full-time	1/1/2024	12/31/2026	1200000	Yes
	CT005	P005	Full Time	12/15/2025	12/30/2025	60000	Yes

1 - 5

Figure 4.5: Player

Cricket Tournament Management

Ticket

Audience

Coach

Match

Schedule

Match Statistics

Attends

Participates

Ranking

Player

Q

Go

Actions

Create

	Player ID ↑	Person ID	Team	Jersey Num	Batting Avg	Batting Style	Role	Bowling Avg	Bowling Style	Nationality
	PL001	P001	Dhaka Dynamites	75	38.5	Left-handed	All-rounder	32.1	Left-arm orthodox	Bangladeshi
	PL002	P002	Chattogram Challengers	15	36.8	Right-handed	Wicket-keeper Batsman	0		Bangladeshi
	PL003	P003	Khulna Tigers	28	40.2	Left-handed	Opening Batsman	0		Bangladeshi
	PL004	P004	Sylhet Strikers	54	25.3	Right-handed	Bowler	28.5	Right-arm medium	Bangladeshi
	PL005	P005	Sylhet Strikers	45	32.2	Left-handed	All Rounder	2.2	Left-handed	
	PL006	P006	East West	10	3	Left hand	Opening Batsman	7	Right hand	Bangladeshi

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Figure 4.6: Ranking

Cricket Tournament Management

Ticket

Audience

Coach

Match

Schedule

Match Statistics

Attends

Participates

Ranking

Ranking

Q

Go

Actions

Create

	Team ID ↑	Net Run Rate	Position	Points	Matches Played	Wins	Losses	Ties
	T001	1.25	2	12	8	6	2	0
	T002	.85	3	10	8	5	3	0
	T003	1.55	1	14	8	7	1	0
	T004	.45	4	4	8	2	6	0
	T005	2.3	3	6	3	2	4	3

1 - 5

Figure 4.7: Participates

Cricket Tournament Management

Ticket

Audience

Coach

Match

Schedule

Match Statistics

Attends

Participates

Participates

Q

Go

Actions

Create

	Tournament ID ↑	Team ID
	TR001	T001
	TR001	T002
	TR001	T004
	TR001	T003
	TR002	T003

1 - 5

Figure 4.8: Attends

Cricket Tournament Management

Ticket

Audience

Coach

Match

Schedule

Match Statistics

Attends

Participates

Attends

Q

Go

Actions

Create

	Audience ID ↑	Match ID
	A001	M001
	A002	M001
	A003	M002
	A004	M003
	A005	M005

1 - 5

Figure 4.9: Match Statistics

Cricket Tournament Management

Ticket

Audience

Coach

Match

Schedule

Match Statistics

Attends

Participates

Match Statistics

Q

Go

Actions

Create

	Stat ID ↑	Match ID	Runs Scored	Catches	Wickets Taken	Overs Bowled	Run Outs	Strike Rate	Economy Rate	Four Hits	Six Hits
	ST001	M001	85	2	3	4	1	145.5	7.25	8	4
	ST002	M002	42	1	0	0	0	120	0	5	1
	ST003	M003	120	0	2	10	0	110.5	5.8	10	3
	ST004	M004	65	1	4	8	1	135	6.5	7	2
	ST005	M005	200	1	2	3	4	54	12	3	3

1 - 5

Figure 4.10: Schedule

Cricket Tournament Management

Tournament

Venue

Ticket

Audience

Coach

Match

Schedule

Match Statistics

Schedule

Q

Go

Actions

Create

	Match ID ↑	Match Time	Match Date	Status
	M001	12/1/2025	12/1/2025	Cancelled
	M002	1/21/2024	1/21/2024	Completed
	M003	1/25/2024	1/25/2024	Completed
	M004	1/26/2024	1/26/2024	Scheduled

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Figure 4.11: Match

Cricket Tournament Management

Home

Person

Team

Tournament

Venue

Ticket

Audience

Coach

Match

Match

Q

Go

Actions

Create

	Match ID ↑	Match Date	Match Type	Team 1 ID	Team 2 ID	Winning Team ID	Tournament ID	Venue ID
	M001	1/20/2024	T20	T001	T002	T001	TR001	V001
	M002	1/21/2024	T20	T003	T004	T003	TR001	V002
	M003	1/25/2024	ODI	T001	T003	T003	TR001	V001
	M004	1/26/2024	T20	T002	T004	T002	TR001	V003
	M005	12/10/2025	T20	T002	T004	T004	TR004	V002
	M006	12/16/2025	ODI	T001	T002	T001	TR003	V002

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Figure 4.12: Coach

Cricket Tournament Management

Home

Person

Team

Tournament

Venue

Ticket

Audience

Coach

Match

Coach

Q

Go

Actions

Create

	Coach ID ↑	Person ID	Team	Certification Level	Experience Years	Speciality	Salary	Nationality
	C001	P002	Dhaka Dynamites	Level 3	15	Batting Coach	300000	Bangladeshi
	C002	P003	Chattogram Challengers	Level 4	12	Head Coach	400000	Bangladeshi
	C003	P004	Khulna Tigers	Level 3	10	Bowling Coach	250000	Bangladeshi
	C004	P001	Sylhet Strikers	Level 4	18	Fielding Coach	350000	Bangladeshi
	C005	P005	Chattogram Challengers	Level 4	3	Bowling Coach	100000	Bangladeshi

1 - 5

Figure 4.13: Audience

Cricket Tournament Management

Home

Person

Team

Tournament

Venue

Ticket

Audience

Coach

Match

Audience

Q

Go

Actions

Create

	Audience ID ↑	Name	Phone	Ticket ID
	A001	Rahim Khan	+8801811223344	A-25
	A002	Fatima Begum	+8801811223345	B-12
	A003	Arif Hossain	+8801811223346	C-08
	A004	Sadia Akter	+8801811223347	V-01
	A005	Pulok	01478981275	B-4021

1 - 5

Figure 4.14: Ticket

Cricket Tournament Management

Home

Person

Team

Tournament

Venue

Ticket

Audience

Coach

Match

Ticket

Q

Go

Actions

Create

	Ticket ID ↑	Price	Seat No	Ticket Type	Purchase Date
	TK001	1500	A-25	VIP	1/10/2024
	TK002	500	B-12	REGULAR	1/11/2024
	TK003	300	C-08	STUDENT	1/12/2024
	TK004	2500	V-01	PREMIUM	1/13/2024
	TK005	60000	B-4021	VIP	12/15/2025

1 - 5

Figure 4.15: Venue

Cricket Tournament Management

Home

Person

Team

Tournament

Venue

Ticket

Audience

Coach

Venue

Q

Go

Actions

Create

	Venue ID ↑	Stadium Name	Country	Food Court	Security
	V001	Sher-e-Bangla National Stadium	Bangladesh	Available	High
	V002	Zahur Ahmed Chowdhury Stadium	Bangladesh	Available	Standard
	V003	Sylhet International Stadium	Bangladesh	Not Available	High
	V004	Bangabandhu National Stadium	Bangladesh	Available	Very High

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Figure 4.16: Tournament

Cricket Tournament Management

[Home](#)
[Person](#)
[Team](#)
[Tournament](#)
[Venue](#)
[Ticket](#)
[Audience](#)
[Coach](#)

Tournament

Q

Go

Actions

Create

	Tournament ID ↑	Name	Host Country	Start Date	End Date	Season Year	Status
	TR001	Bangladesh Premier League	Bangladesh	1/5/2024	2/15/2024	2024	Completed
	TR002	Dhaka Premier Division	Bangladesh	3/1/2024	5/30/2024	2024	Ongoing
	TR003	Bangladesh T20 Cup	Bangladesh	11/10/2023	12/20/2023	2023	Completed
	TR004	National Cricket League	Bangladesh	10/1/2024	12/31/2024	2024	Scheduled

1 - 4

Figure 4.17: Team

Cricket Tournament Management

auma v

[Home](#)
[Person](#)
[Team](#)
[Tournament](#)
[Venue](#)
[Ticket](#)
[Audience](#)
[Coach](#)
[Match](#)

Team

	Team ID ↕	Team Name	Country	Owner Name	Budget	Title Sponsor
	T001	Dhaka Dynamites	Bangladesh	Kazi Inam Ahmed	50000000	Daraz
	T002	Chattogram Challengers	Bangladesh	Nazmul Hasan	45000000	Akash DTH
	T003	Khulna Tigers	Bangladesh	Mizanur Rahman	42000000	Bashundhara
	T004	Sylhet Strikers	Bangladesh	Abdur Razzak	40000000	Prime Bank
	T005	East West	Bangladesh	Aurna	560000	Robi

1 - 5

Figure 4.18: Person

Cricket Tournament Management

Home
 Person
 Team
 Tournament
 Venue
 Ticket
 Audience
 Coach
 Match

Person

Go
Actions ▾
Create

	Person ID ↑	Name	Date of Birth	Age	Street	City	Postal Code	Country	Email	Gender	Phone Number
	P001	Shakib Al Hasan	3/24/1987	38	Magura Sadar	Magura	7600	Bangladesh	shakib@cricket.bd	Male	+8801711000001
	P002	Mushfiqur Rahim	5/9/1988	37	Bogura Sadar	Bogura	5800	Bangladesh	mushfiq@cricket.bd	Male	+8801711000002
	P003	Tamim Iqbal	3/20/1989	36	Chittagong Port	Chittagong	4100	Bangladesh	tamim@cricket.bd	Male	+8801711000003
	P004	Nigar Sultana	8/15/1993	32	Rajshahi City	Rajshahi	6000	Bangladesh	nigar@cricket.bd	Female	+8801711000004
	P005	Nusrat	12/1/2003	22	Sheikhdhi	Dhaka	1203	Bangladesh	nusrat@cricket.bd	Female	01533359793
	P006	Zeba	12/1/2025	0	Road 4	Dhaka	1205	Bangladesh	zeba@gmail.com	Female	01236786393

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5. Reports based on Multi-Table and Aggregate Queries

Page Number and Name	Query Type	Report description
Page No – 34, Sponsored By	Multi-Table	Shows which sponsors are funding which tournaments
Page No – 24, Participates	Multi-Table	Tracks team participation across different tournaments
Page No – 22, Attends	Multi-Table	Records audience attendance at cricket matches
Page No – 42, Total audience per match	Aggregate Queries	Total audience per match
Page No – 44, Players earning more than the team-wise average salary	Aggregate Queries	Players earning more than the team-wise average salary
Page No – 64, Tournament with the highest number of participating teams	Aggregate Queries	Tournament with the highest number of participating teams
Page No – 46, Teams Participating in Tournaments with High Audience Turnout	Aggregate Queries	Teams Participating in Tournaments with High Audience Turnout
Page No – 48, Get the average ticket price for each type of ticket (REGULAR, VIP, etc.) and the total number of tickets sold	Aggregate Queries	Get the average ticket price for each type of ticket (REGULAR, VIP, etc.) and the total number of tickets sold
Page No – 50, Find the top 3 venues that hosted the most matches, sorted by match count	Aggregate Queries	Find the top 3 venues that hosted the most matches, sorted by match count
Page No – 52, Tournaments That Have Matches Scheduled but Not Completed	Aggregate Queries	Tournaments That Have Matches Scheduled but Not Completed
Page No – 54, Cities That Have More Than One Registered Person	Aggregate Queries	Cities That Have More Than One Registered Person
Page No – 56, Players Whose Contract Salary Is the Maximum in Their Team	Aggregate Queries	Players Whose Contract Salary Is the Maximum in Their Team

Page No – 58, Teams With More Wins Than Losses	Aggregate Queries	Teams With More Wins Than Losses
Page No – 60, Ticket Types That Are Sold More Than the Average Ticket Sales	Aggregate Queries	Ticket Types That Are Sold More Than the Average Ticket Sales
Page No – 62, List of Coaches Whose Name Starts with ‘A’ and Salary Above Average	Aggregate Queries	List of Coaches Whose Name Starts with ‘A’ and Salary Above Average

Aggregate Queries:

Figure 5.1: Total audience per match

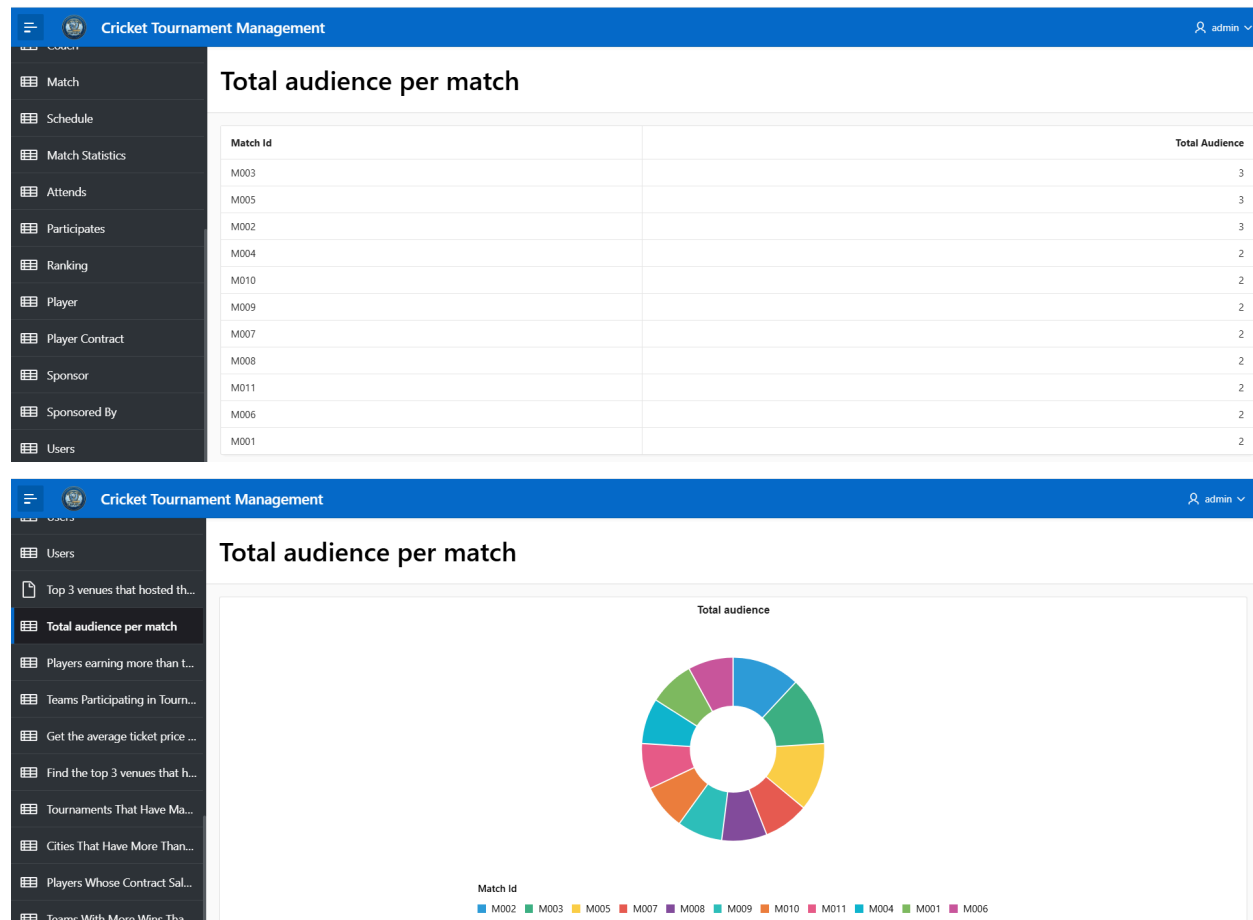


Figure 5.2: Players earning more than the team-wise average salary

Cricket Tournament Management			admin
Schedule	Players earning more than the team-wise average salary		
Match Statistics			
Attends			
Participates			
Player Id	Salary	Team Id	
PL004	1200000	T004	

Figure 5.3: Tournament with the highest number of participating teams

Cricket Tournament Management			admin
Home	Tournament(s) with the highest number of participating teams		
Tournament(s) with the highest ...			
Person			
Team			
Tournament Id	Team Count		
TR001	8		

Figure 5.4: Teams Participating in Tournaments with High Audience Turnout

Cricket Tournament Management			admin
Match Statistics	Teams Participating in Tournaments with High Audience Turnout		
Attends			
Participates			
Ranking			
Player			
Player Contract			
Sponsor			
Sponsored By			
Home			
Team Name			
Barisal Bulls			
Chattogram Challengers			
Comilla Victorians			
Dhaka Dynamites			
East West			
Khulna Tigers			
Rajshahi Kings			
Sylhet Strikers			

Figure 5.5: Get the average ticket price for each type of ticket (REGULAR, VIP, etc.) and the total number of tickets sold

Cricket Tournament Management			admin
Attends	Get the average ticket price for each type of ticket (REGULAR, VIP, etc.) and the total number of tickets sold		
Participates			
Ranking			
Player			
Player Contract			
Sponsor			
Sponsored By			
Home			
Ticket Type	Average Price	Total Tickets Sold	
REGULAR	650	8	
VIP	11566.67	6	
STUDENT	361.67	6	
PREMIUM	2875	4	

Figure 5.6: Find the top 3 venues that hosted the most matches, sorted by match count

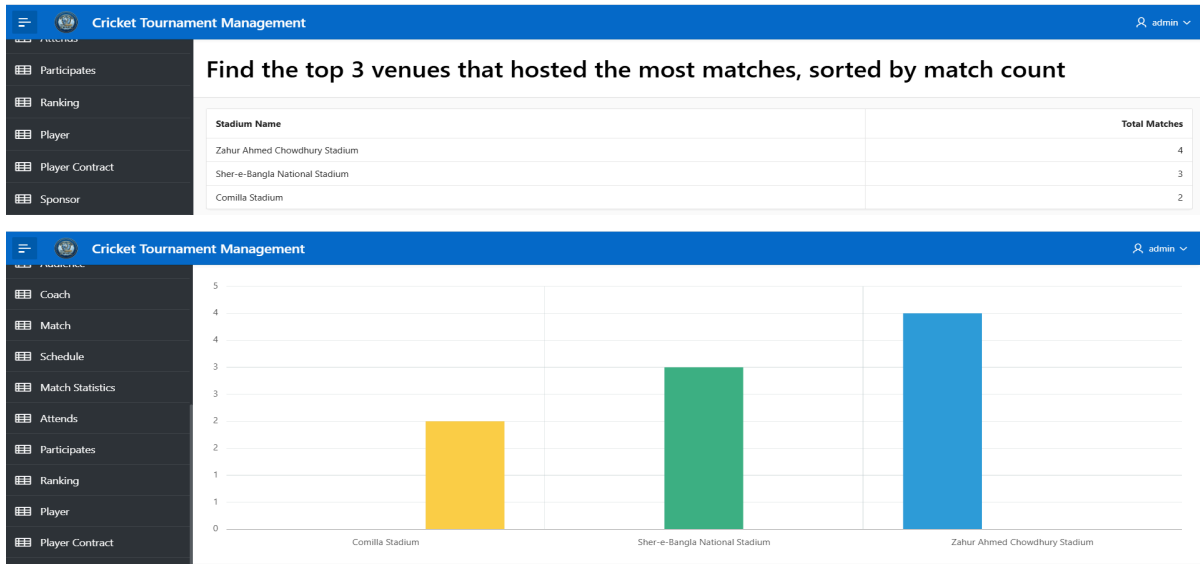


Figure 5.7 : Tournaments That Have Matches Scheduled but Not Completed

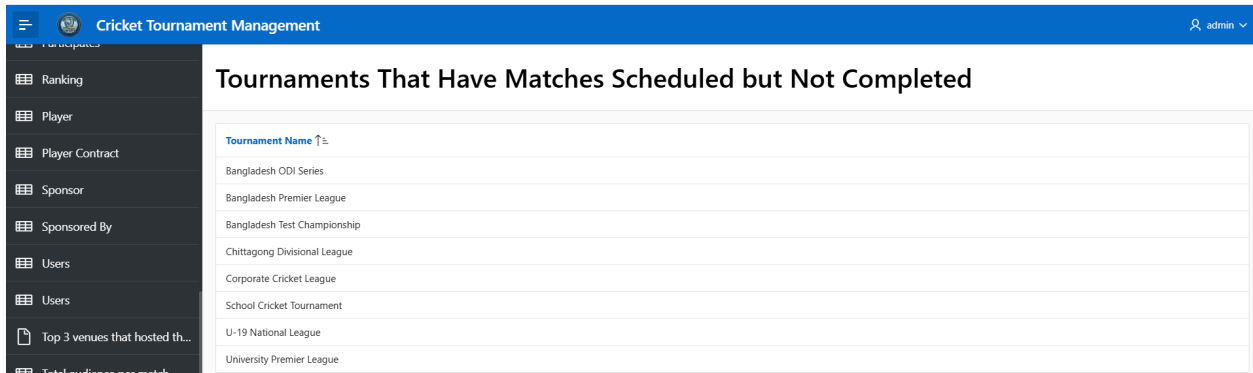


Figure 5.8: Cities That Have More Than One Registered Person

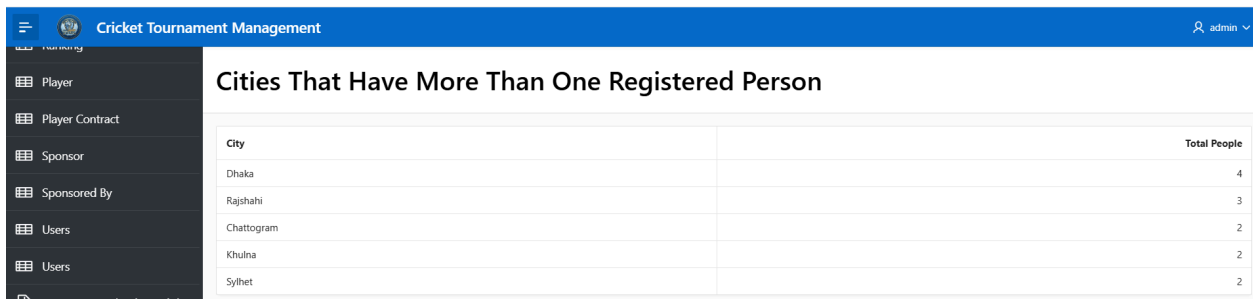


Figure 5.9: Players Whose Contract Salary Is the Maximum in Their Team

Cricket Tournament Management			
admin			
Players Whose Contract Salary Is the Maximum in Their Team			
Player Id	Salary	Team Id	
PL001	2000000	T001	
PL002	1800000	T002	
PL003	1500000	T003	
PL004	1200000	T004	
PL006	1400000	T005	
PL007	1100000	T007	
PL008	1000000	T008	
PL009	1300000	T009	

Figure 5.10: Teams With More Wins Than Losses

Cricket Tournament Management			
admin			
Teams With More Wins Than Losses			
Team Name	Wins	Losses	
Khulna Tigers	7	1	
Bangladesh National Team	6	0	
Dhaka Dynamites	6	2	
Chattogram Challengers	5	3	
Dhaka Gladiators	5	0	
Bangladesh Women Team	4	2	
Chattogram Vikings	4	1	

Figure 5.11: Ticket Types That Are Sold More Than the Average Ticket Sales

Cricket Tournament Management		
admin		
Ticket Types That Are Sold More Than the Average Ticket Sales		
Ticket Type	Total Sold	
REGULAR	8	
VIP	6	
STUDENT	6	
PREMIUM	4	

Figure 5.12: List of Coaches Whose Name Starts with ‘A’ and Salary Above Average

Cricket Tournament Management		
admin		
List of Coaches Whose Name Starts with ‘A’ and Salary Above Average		
Name	Salary	Team Name
Afif Hossain	800000	Bangladesh National Team

6. Authorization and User Role Assignment

User Role Assignment

User name	Password	Role
Admin	Admin123	Admin
Coach	Coach123	Coach
Player	Player123	Player
User	user123	User

Figure 6.1: User Report showing role assignments

The screenshot displays the 'Cricket Tournament Management' application interface. On the left is a dark sidebar with a menu containing items like Home, Person, Team, Tournament, Venue, Ticket, Audience, Coach, Match, Schedule, Match Statistics, Attends, Participates, Ranking, Player, Player Contract, Sponsor, Sponsored By, and Users. The main content area is titled 'Users' and features a table with columns: User ID, User Name, User Pass, User Role, and User Activated. The table lists four users: Admin (ID 1), Coach (ID 2), Player (ID 3), and User (ID 4). Each user row has a blue checkmark icon in the first column. A 'Create' button is located in the top right corner of the table area. The footer of the application shows 'Release 1.0'.

User ID	User Name	User Pass	User Role	User Activated
1	Admin	Admin123	admin	1
2	Coach	Coach123	coach	1
3	Player	Player123	player	1
4	User	user123	user	1

7. Concluding Remarks

The Cricket League/Tournament Management project provides a well-structured system for managing cricket tournaments. It uses a clean database schema with surrogate primary keys to reduce redundancy and improve data integrity. Key features include automated age calculation for players, strong data validation through constraints, and efficient management of relationships between teams, players, and matches. The system supports role-based access for admins, coaches, players, and users, ensuring secure and organized operations. Advanced reports and queries track tournament participation, audience attendance, and player performance. Overall, this project is a reliable and efficient solution for managing cricket tournaments.